

H2-SHIFT Round 2: three-minute video pitch script and recording storyboard

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Length: 379 spoken words; planned duration 176 seconds (2 min 56 s) including pauses, about 129 words per minute. Read aloud once with a stopwatch; if it runs over 178 seconds, drop the second sentence of segment 3 (about 20 words).

Script

Slide 1 (12 s): Hello, we are Team Wookies from NIT Warangal. In Round 1 we proposed H2-SHIFT: convert India's CNG cars to hydrogen instead of building new ones. Round 2 tests that idea against the evidence.

Slide 2 (22 s): India has 8.2 million CNG vehicles and Maruti sells seven lakh CNG cars a year. Our first customer is a captive institutional fleet in Faridabad, running Dzire Tour S CNG cars 150 to 200 kilometres a day beside India's only car-capable hydrogen dispensing point.

Slide 3 (18 s): The route is hydrogen combustion with port injection and lean burn. A fuel-cell conversion would cost as much as a new BEV, and global experience shows depot fleets survive while retail networks retreat.

Slide 4 (24 s): We replace the CNG cylinder with a 60-litre composite hydrogen cylinder, add a regulator, injectors, a dual-map ECU and twin leak sensors, and keep petrol as automatic backup. One fill gives about 60 kilometres, two depot fills about 120. Power drops by roughly half: a city car.

Slide 5 (19 s): Safety is engineering we know: a sealed, vented boot, thermal relief, valve closure within a second, lean burn with a NOx trap. The real gap is approval: no route exists today for a hydrogen retrofit of a car. That is our first gate.

Slide 6 (27 s): Now the honest economics. At 800 rupees a kilogram in 2028 the converted car costs about 14 rupees a kilometre, against 5 on CNG and 5.5 for a compact BEV including purchase. Fuel parity with CNG needs hydrogen below 190 rupees a kilogram, cheaper than any 2030 forecast we found. The retrofit is a decarbonisation cost, not a saving.

Slide 7 (15 s): A 30-car pilot needs about 60 kilograms of hydrogen a day, delivered by tube trailer to a shared dispensing point in Faridabad: roughly 1,400 rupees a kilogram today, under 600 at a 2030 hub.

Slide 8 (20 s): So we recommend an option, not a launch: four gated phases to 2030, 37.6 crore rupees, each ending in a proceed, redesign or stop decision, funded jointly by Suzuki, the MNRE pilot scheme, a fuel partner and the fleet.

Slide 9 (19 s): The impact is real but small by 2030: about 1,400 cars, 3,600 tonnes of CO₂ a year, and India's first approval route for hydrogen retrofits. We ask for 0.9 crore for Phase 0: it buys the facts that decide whether the rest is worth spending. Thank you.

Timing table

Time window	Slide	Spoken words	Presenter action	Visual emphasis
0:00 to 0:12	1	33	Look to camera, calm opening	Title slide unchanged
0:12 to 0:34	2	44	Point to the funnel column	Log-scale funnel bars, 8.2 million to about 1,400; conditions table
0:34 to 0:52	3	33	Gesture across the comparison table	Comparison matrix, highlighted H2-ICE column; deployment status ladder
0:52 to 1:16	4	47	Trace the flow in the architecture diagram	Sedan schematic with added, retained, modified, removed legend; spec strip 1.12 kg, 61 km, 42 PS
1:16 to 1:35	5	43	Slow down on 'no route exists'	Hazard matrix; approval sequence with established, unresolved and gap chips
1:35 to 2:02	6	59	Pause after 'not a saving'	Stacked five-year cost chart; conversion cost breakdown; hydrogen price heat grid
2:02 to 2:17	7	34	Point to the stacked bars	Supply-to-fleet flow; utilisation versus price curves with parity lines

2:17 to 2:37	8	39	Sweep along the chevrons	Gantt lanes with gate diamonds; gate table; funding bar
2:37 to 2:56	9	47	Close to camera, hold two seconds	Emissions bars; pilot scorecard; decision panel
Total		379	176 s	

Recording layout and export settings

- Layout: split screen. The slide occupies the left 76 percent of the frame; the presenter occupies the right column (about 24 percent) with a plain background, so no chart label or table cell is covered. If the organiser's picture-in-picture layout is preferred (presenter bottom-right, about 24 percent wide and 38 percent tall as in the reference image), the deck was built so that footnotes sit bottom-left and only source lines fall under that zone.
- Both presenter and solution must be visible throughout; keep the presenter on screen during the title slide and the closing banner.
- Slides: use the PDF export (identical fonts on every machine); advance exactly at the timing-table boundaries.
- Video: 1920 x 1080, 25 or 30 fps, H.264 (MP4, High profile), 2-pass at 1.8 to 2.2 Mbit/s video; audio AAC 128 kbit/s stereo, 48 kHz; expected file size 40 to 48 MB for 176 s; verify below 50 MB before upload and reduce the video bitrate to 1.6 Mbit/s if needed.
- Audio: lapel or USB microphone 30 cm from the mouth, quiet room, record a 5-second silence for noise removal; peak level minus 6 dB.
- Check after export: actual duration between 170 and 180 s, file size below 50 MB, slide text legible at 100 percent, audio in sync at the slide changes on slides 4 and 6.
- Presenter recording is the remaining input: no video file has been produced; the script, storyboard, layout and export settings are complete.

Do-not-say list (keeps the video consistent with the deck)

- No 'approved', 'fully safe', 'zero emission' (hydrogen combustion emits NOx), 'partnership' or 'letter of intent'.
- No numbers that are not on the slides; the pitch uses only slide figures rounded for speech.
- Say 'proposed partner' for IOCL, ARAI, fleets and Suzuki R&D Center India.